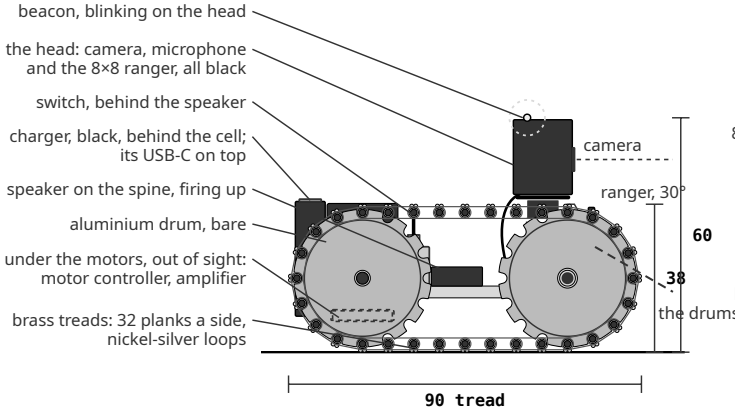
Mars rover: how the finished thing looks

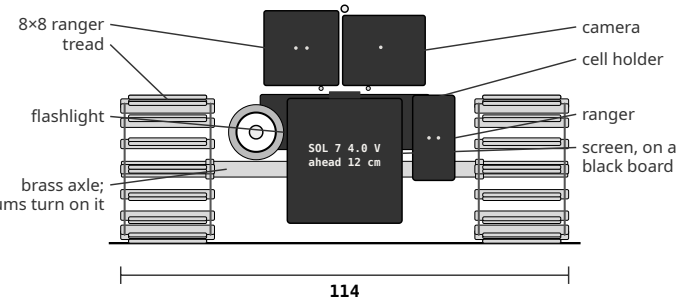
A wide little tank on a brass spine: brass treads on bare aluminium drums, six brass tubes, every part in its own colour (brass, aluminium, black), the blue and green boards out of sight. 114 × 90 × 60 mm, about 332 g; one scale.

Side view, from the rover's right

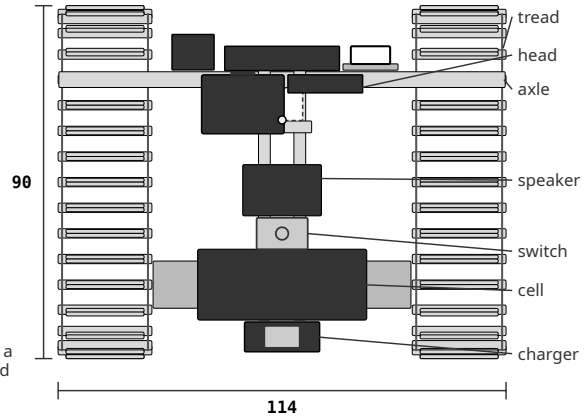


Front view

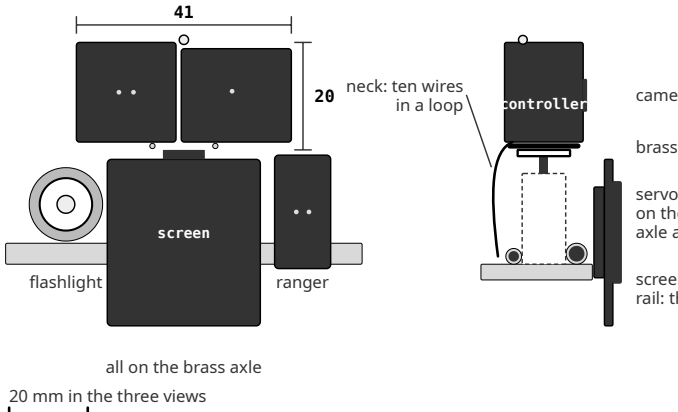
the rover's right is on your left



Top view, front up

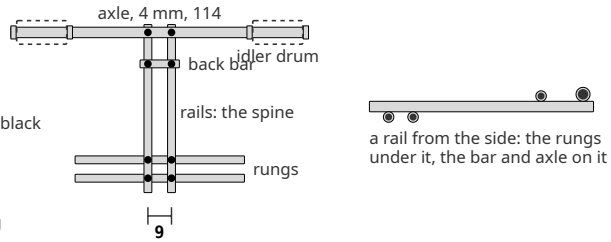


The face and a cut through the servo, at 3.6 px per mm



20 mm in the three views

The frame: six brass tubes, eight laps



a rail from the side: the rungs under it, the bar and axle on it

3 mm tube: two rails 64, two rungs 65, the back bar 15 mm
4 mm tube: the axle, 114 mm; 5 mm: four collars, 2 mm each
Six straight tubes, no bends; 8 laps soldered. The frame is the ground. The rails' open ends carry the wires inside: the bus and BAT+ in the left, the sound in the right.

Key numbers

114 × 90 × 60 mm	wide, long, tall
332 g	centre 21 mm up; tips 69°, 50° back
8.3 cm/s	on the desk; turns on the spot
6 tubes	the whole frame, 8 laps, and the ground
head 41 × 20 × 16	turns 120°; it hears and sees
64 planks	6.5 mm apart on 16-seat drums
1.9 h / 3.1 h	exploring / talking, on one 16340
2.2 h	to charge, USB-C at 500 mA
5.1 mm	clearance: the screen's edge

Brass, aluminium and black

Brass treads and frame, bare aluminium drums; black head, screen, ranger, charger, cell holder and speaker, as they are sold. No paint, no skins: the blue and green boards hang under the motors, out of sight. White light, one warm on the head.

Rev K for approval, 2026-11-10: brass frame, no paint, a voice. Rev J approved 2026-11-09; Rev H 2026-11-08; Rev A 2026-11-06. Nothing cut or soldered yet. Caliper the drums, the boards and the servo first.